

FRACMATH: A vectorized MATLAB framework for continuum damage mechanics with crack band regularization

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Abstract

FRACMATH is an inspectable, vectorized MATLAB implementation of isotropic damage with direction-dependent crack-band regularization. Its code links projected width, constitutive history, sparse assembly and solution. Material-point tests verify strain mapping, irreversible damage and fracture-energy calibration. Mesh/increment studies compare regularized and fixed softening laws. Ninety observations of thirty matched computing configurations assess MATLAB CPU, hybrid GPU and Abaqus SMP configurations. The hybrid GPU shows no speed advantage on the tested workstation. A geometry-matched three-dimensional pure-tension comparison supplies experimental error measures; mixed-mode and torsion images illustrate workflows. Source, histories, diagnostics and reproduction scripts accompany the paper.

Keywords: continuum damage mechanics, crack-band regularization, MATLAB, hybrid GPU, Abaqus UMAT, quasibrittle fracture

Required Metadata

Current code version

Nr.	Code metadata description	Value
C1	Current code version	v1.1.1, commit f4d208e
C2	Permanent link to code/repository used for this code version	Source and study data: immutable GitHub commit ; version archive DOI .
C3	Legal code license	MIT License
C4	Code versioning system used	git
C5	Software code languages, tools and services used	MATLAB; Fortran UMAT; Abaqus/Standard; Python analysis scripts
C6	Compilation requirements, operating environments and dependencies	MATLAB R2024b; Abaqus/Standard 2024 with Intel Fortran; Python 3 with NumPy, SciPy, Matplotlib, Pillow, and psutil; Parallel Computing Toolbox and a compatible CUDA GPU for the optional hybrid backend. Verified on Windows 11.
C7	If available, link to developer documentation/manual	Beginner guide ; implementation walkthrough and reproduction guides
C8	Support email for questions	mpathirage@unm.edu

Table 1: Code metadata.

Motivation and significance

FRACMATH is intended for researchers and students who need to inspect the numerical path from a strain field to a damage field and a global response. The constitutive law and crack-band principle

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are established methods [1, 2, 3]. The useful software question is whether the implementation makes its assumptions, update sequence, regularization length, and numerical limitations visible enough to check and modify. The package therefore includes element-level routines, a 2D bending case, two 3D examples, plotting scripts, and an Abaqus UMAT for independent comparison. The controlled study examines how element-size compensation affects structural response and dissipation, and how increment size and sparse linear algebra limit the numerical workflow.

Related software and scope

General-purpose finite element systems such as Abaqus [4] and OOFEM [5] provide broader element and analysis capabilities. CALFEM [6] provides an accessible MATLAB finite element teaching environment. FRACMATH offers a narrower, code-readable example of direction-dependent local crack-band damage in 2D and 3D. It is designed for teaching, replication, and testing method changes.

Software description

A scalar damage variable $\omega \in [0, 1]$ scales the elastic stress: $\boldsymbol{\sigma} = (1 - \omega)\mathbb{C}_0 : \boldsymbol{\varepsilon}$. The history variable $\kappa_{n+1} = \max(\kappa_n, \tilde{\varepsilon}_{n+1})$ retains maximum equivalent strain. Committing $\omega_{n+1} = \max(\omega_n, \omega_{\text{trial}})$ enforces damage irreversibility when the projected width changes. The modified von Mises measure is

$$\tilde{\varepsilon} = \frac{k-1}{2k(1-2\nu)}I_1 + \frac{1}{2k}\sqrt{\left(\frac{k-1}{1-2\nu}\right)^2 I_1^2 + \frac{12k}{(1+\nu)^2}J_2}, \quad J_2 = \frac{(e_1 - e_2)^2 + (e_2 - e_3)^2 + (e_3 - e_1)^2}{6}, \quad (1)$$

where $k = f_c/f_t$. For the 2D plane-stress element, $e_3 = -\nu(e_1 + e_2)/(1 - \nu)$. Both the MATLAB and UMAT implementations use Eq. (1).

The damage law is

$$\omega(\kappa) = 1 - \frac{\kappa_0}{\kappa} \exp\left[-\frac{\kappa - \kappa_0}{\varepsilon_f - \kappa_0}\right], \quad \kappa_0 = f_t/E, \quad \varepsilon_f = \frac{\kappa_0}{2} + \frac{G_F}{hf_t}. \quad (2)$$

The Oliver projected width for a three-node triangle is

$$h(\mathbf{n}) = \frac{2}{\sum_{a=1}^3 |\nabla N_a \cdot \mathbf{n}|}, \quad (3)$$

with $\mathbf{n}$ along maximum principal strain. The 3D tetrahedral implementation uses the analogous projected width. Its exponential law uses $\beta = f_t h/G_F$ and $\omega = 1 - (\kappa_0/\kappa) \exp[-\beta(\kappa - \kappa_0)]$, capped at 0.999999. Here G_F calibrates post-peak work; Eq. (2) calibrates total uniaxial work in 2D. These are local regularization formulas; the material-point energy check below does not by itself prove structural mesh objectivity. The pure-tension 3D implementation retains both κ and ω ; accepted increments commit their nondecreasing histories, while rejected trials leave the committed state unchanged.

Numerical workflow and code map

Students open `start_here.m`, edit the visible settings and run the supplied mesh. The beginner guide explains units, array dimensions and the function sequence. The entry script calls the benchmark solver directly. A headless 10,000-step execution of the entry exactly reproduces all saved benchmark response, state and snapshot arrays.

The 2D driver `solver_main_3pb.m` reads the mesh and support/loading sets, precomputes elastic element matrices and shape-function gradients, and advances prescribed displacement. At each increment it (1) assembles the secant stiffness from the previous damage field, (2) factors its free block with `decomposition`, (3) solves equilibrium for the frozen damage state, (4) obtains element strains with `pagetimes`, updates h , κ , and ω in `damage_update`, and (5) reassembles

the damaged stiffness for reaction forces and the next increment. The factorization helper first attempts Cholesky and falls back to LU if that attempt fails. It is reused only within a displacement increment. There is no outer equilibrium iteration after the damage update, so this is a sequential secant scheme rather than a fully converged Newton solve. The post-update free-DOF residual is exported for each increment.

For example, the projected-width calculation acts on each triangle’s three gradient pairs. With `gradN`, `nx` and `ny` denoting those gradients and principal-direction components, its MATLAB form is

```
projected = gradN(:,1:2:end).*nx + gradN(:,2:2:end).*ny;
h = max(2./max(sum(abs(projected),2),1e-14),1e-12);
```

This width enters Eq. (2); `kappa` and `omega` retain the irreversible element histories.

The optional Abaqus path uses `run_3pb_abaqus_OLIVER_T3_FAST.py` to write a table of T3 shape-function gradients. The Fortran `cdm_umat_2d_OLIVER_T3_FAST.for` reads that table to compute Eq. (3); it does not use `CELENT` when the table is present. The UMAT supplies the degraded elastic secant matrix as `DDSDDE`, rather than a consistent derivative of the evolving damage law. Its approximate tangent requires reporting equilibrium iteration counts with timings. `UEXTERNALDB` initializes the gradient table before SMP workers start; material calls read its saved arrays, and missing gradients terminate the job.

The optional 2D hybrid GPU backend keeps strain operators and reference element matrices on the device. A fused `gpuArray.arrayfun` kernel evaluates projected width, equivalent strain, and irreversible history. Results and stiffness values return to CPU sparse assembly and factorization. Timed operations include transfers and synchronization.

Material-point verification

A one-element uniaxial test checks $\tilde{\varepsilon} = \varepsilon$ in tension and $\tilde{\varepsilon} = |\varepsilon|/k$ in compression, monotone damage under unload/reload, and $h \int \sigma d\varepsilon \approx G_F$ for four prescribed projected widths. Numerical integration to a large terminal strain gives the following results for $G_F = 0.090$ N/mm. A standalone driver linked to the unchanged Fortran UMAT repeats the strain-mapping, projected-width, energy, and unload/reload assertions; its energy results agree with the MATLAB values at the precision shown. Sixteen additional UMAT checks combine these four triangle widths with loading directions of 15° , 45° , 75° , and 90° . Projected widths agree with Eq. (3), and the largest absolute relative energy error is 1.78×10^{-6} . These are local implementation checks, not evidence of structural orientation independence. An additional 712-case audit compares the actual UMAT with an independent tensor-eigensolve calculation and the MATLAB damage functions. Strain splits, multi-axial states, irreversible history, projected width, stress and the documented secant matrix pass declared absolute/relative tolerances of 2×10^{-10} . This does not establish a consistent damage tangent.

Table 2: Material-point softening-energy checks. Oliver relative error is against $G_F = 0.090$ N/mm. The fixed-law integral targets $G_F h/h_{\text{ref}}$ with $h_{\text{ref}} = 1.25$ mm. These are local checks.

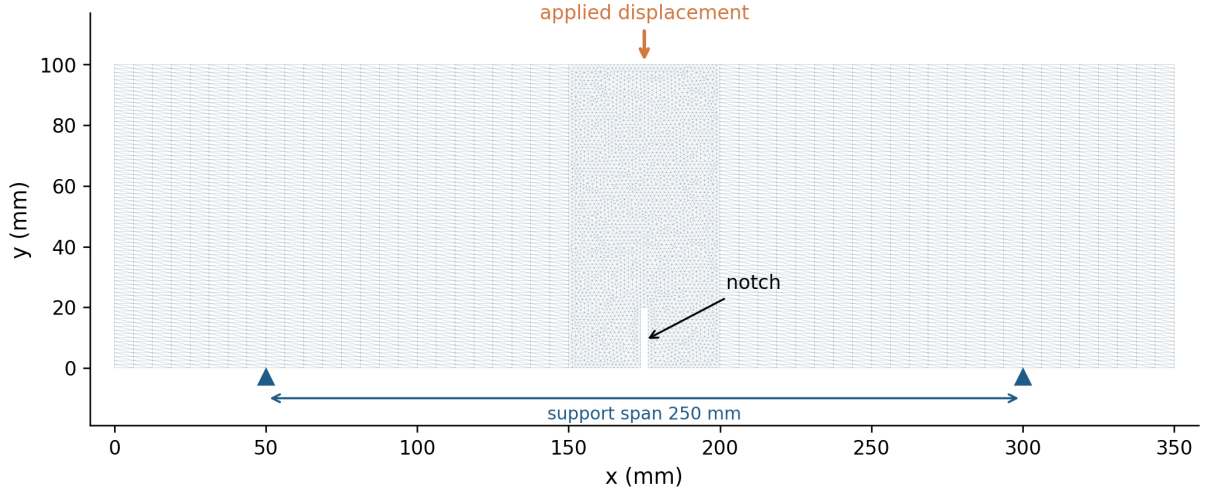
h (mm)	Oliver integral (N/mm)	Relative error	Fixed-law integral (N/mm)
0.5	0.090000160	1.78×10^{-6}	0.036000064
1.0	0.090000160	1.77×10^{-6}	0.072000128
2.0	0.090000159	1.77×10^{-6}	0.144000255
4.0	0.090000159	1.76×10^{-6}	0.288000511

Illustrative examples

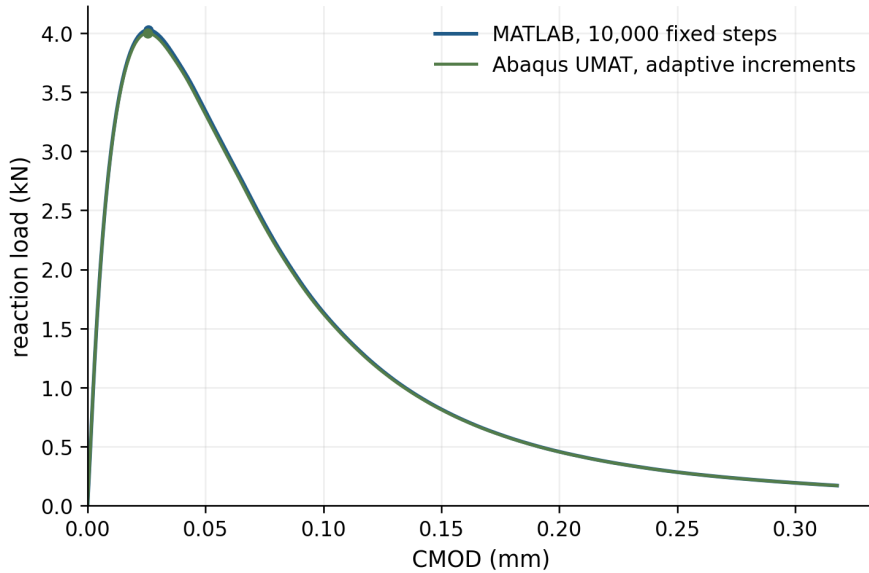
Benchmark 1: Notched 2D three-point bending

The Grégoire beam [7] has depth $D = 100$ mm, support span $S = 250$ mm, overall length 350 mm, thickness 50 mm, and a midspan notch of depth 20 mm and width 2.5 mm. Concrete

parameters are $E = 37,000$ MPa, $\nu = 0.20$, $f_t = 3.5$ MPa, $f_c = 35$ MPa, and $G_F = 0.090$ N/mm. The 14,268 CPS3-element mesh has 7,319 nodes. The left support is pinned, the right is a roller, and a prescribed vertical displacement reaches -0.2 mm. CMOD is the mean horizontal displacement difference across the notch mouth. Figure 1 uses the mesh drawing: the 250 mm dimension denotes support span, while 350 mm is the overall length.



(a) Geometry, mesh, supports, and imposed displacement.



(b) MATLAB 10,000-step response and Abaqus adaptive-increment response.

Figure 1: 2D three-point bending case.

The comparison in Fig. 1 uses the 10,000-step MATLAB response and the Abaqus UMAT response. The separate increment-sensitivity check found that the 1,000-step MATLAB run peaks at 4.26 kN (CMOD 0.02835 mm, step 174), while the 10,000-step run peaks at 4.03 kN (CMOD 0.02579 mm, step 1612). This 5.9% change in peak load shows that increment size matters for the sequential scheme; these results are numerical examples, not an increment-converged validation against experiment. The normalized post-update free-DOF residual at the peak is 0.016 for 1,000 steps and 0.002 for 10,000 steps. A single-thread Abaqus/Standard 2024 UMAT run produced a peak of 4.00 kN at CMOD 0.02540 mm. The 10,000-step MATLAB peak differs by 0.68% in load and 1.51% in peak CMOD. The close response is a code-to-code check, not experimental validation.

The completed Abaqus .msg reports 1,136 accepted increments, 4,817 equilibrium iterations/solver passes, 4,799 matrix decompositions, one cutback, and 45 acceptances under the alternate force tolerance. Its summed per-pass sparse-solver elapsed time is 100.2 s out of 832 s reported wall clock. The remaining wall time includes element/UMAT operations, assembly, con-

vergence and output, which the `.msg` does not separate. Thus no attribution of the remainder to the UMAT or speed ranking between codes is supported. The MATLAB run used 10,000 fixed increments, while Abaqus used automatic increments, so the two wall-clock totals are not a controlled benchmark. The repository includes a parser that reproduces these counts and timings from the archived `.msg` and MATLAB timing file. MATLAB runs export per-step component timings, iteration counts, and the post-damage equilibrium residual. The implementation walkthrough maps these measurements to the corresponding solver functions. The 10,000-step archive supplies aggregate timing and response data; the controlled mesh and hardware archives also supply per-step diagnostics.

Figure 2 compares damage fields at peak and at the first saved post-peak state above CMOD 0.30 mm. The Abaqus field output is taken from the nearest stored frame, while the MATLAB panels use the state arrays.

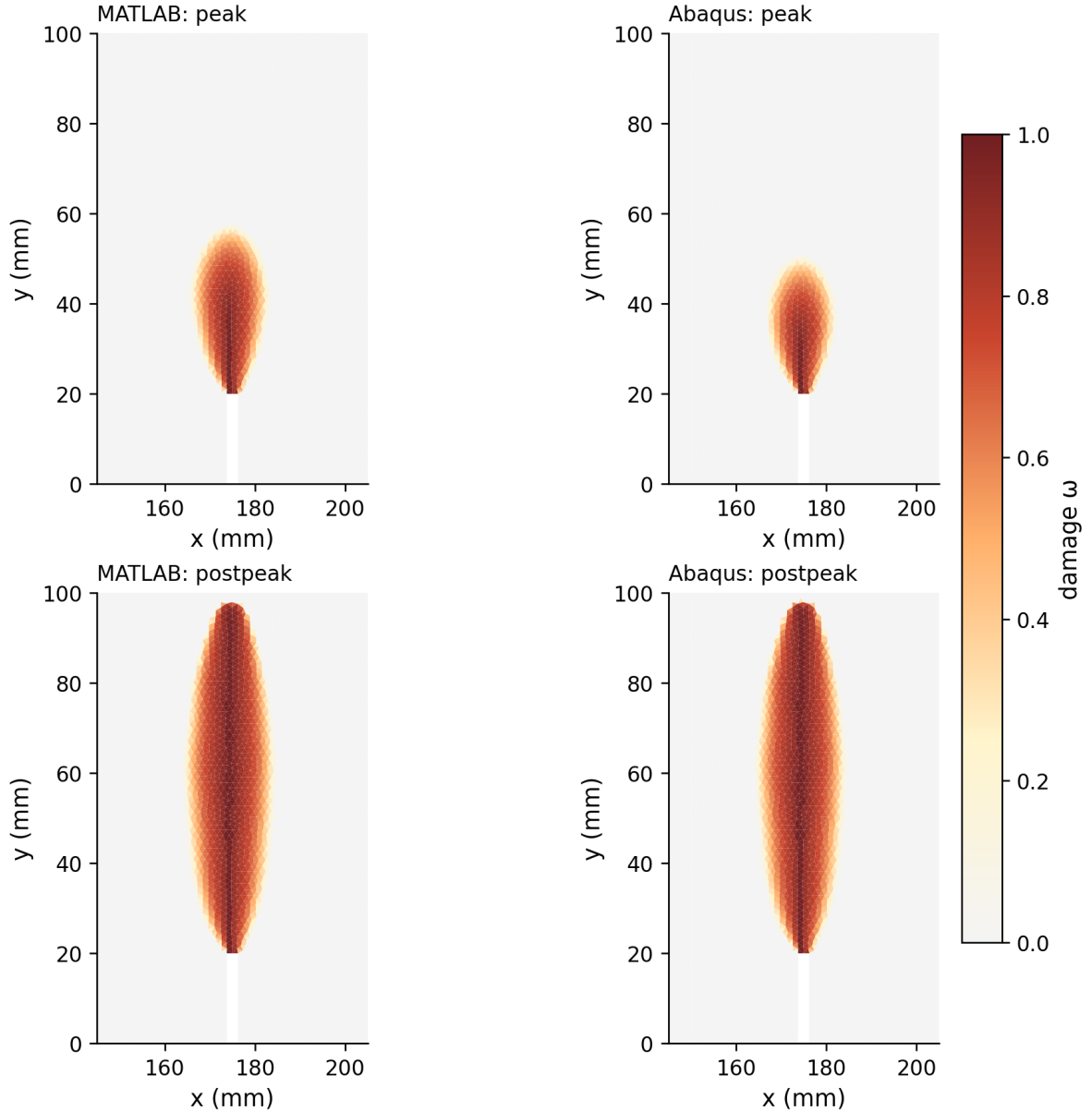


Figure 2: Damage field ω in the notch region on the undeformed CPS3 mesh. Left: MATLAB; right: Abaqus/UMAT. Top: near peak load; bottom: post-peak state. Abaqus fields are the closest stored frames to the indicated response points.

The 10,000-step headless run used MATLAB R2024b, one computational thread, Windows 11, an Intel Core i9-13900 and 64 GB RAM. Tables 3–4 distinguish MATLAB assembly, factorization,

damage and solves from Abaqus sparse-solver passes and unallocated elapsed time. Different algorithms, convergence quality and timing scopes qualify cross-program comparisons.

Table 3: Measured MATLAB 2D timing for the 10,000-step headless run. Other is total wall time minus the instrumented scopes.

Component	Time (s)	Share
Assembly	174.91	22.3%
Sparse factorization	519.54	66.1%
Damage update	17.95	2.3%
Free-block solve	71.52	9.1%
Other	1.78	0.2%
Total	785.70	100%

Table 4: Abaqus/Standard elapsed time extracted from the job’s `.msg`. The remaining time cannot be partitioned into UMAT, assembly, convergence, output, or overhead using this record.

Component	Time (s)	Share
Summed sparse-solver passes (4,817)	100.15	12.0%
Remaining wall time (unallocated)	731.85	88.0%
Total Abaqus wall clock	832.00	100%

Size, mesh and computing configurations

Thirty configurations, each measured three times, combine depths of 100/200 mm, three meshes, and MATLAB CPU1/CPU8/hybrid GPU or Abaqus CPU1/CPU8. All dimensions scale by two; final displacement is $-0.1/-0.2$ mm. Absolute global/notch seeds are 12.5/2.5, 6.25/1.25 and 4.6875/0.9375 mm; each computing comparison uses identical mesh files. The largest mesh has 100,104 elements and 101,088 DOFs. MATLAB uses 2,000 fixed increments; Abaqus uses maximum increments of 1/2,000, with SDV every 100 accepted increments. The two finest large-specimen Abaqus meshes use limits of 80 equilibrium iterations and 12 attempts, retaining default convergence tolerances; a failed default-control attempt is archived. Fresh processes run sequentially; the third observation reverses the computing-configuration order. MATLAB CPU8 limits numerical-library threads; the hybrid backend uses an NVIDIA RTX A2000 12GB in double precision.

All hardware-response checks pass: MATLAB load-history differences are below 1.76×10^{-11} of peak load and final-damage differences below 1.34×10^{-9} ; Abaqus pairs have identical curves at CSV precision and identical iteration counts. MATLAB CPU8 and hybrid GPU are slower than CPU1 in all six cases. CPU1 factorization occupies 67–76% of loop time. Abaqus SMP speed ratios range from 0.68 to 2.90; the largest mesh reaches 2.90. Abaqus CPU1 sparse-solver passes occupy 9.5–14.4% of analysis wall time; the remainder is unallocated. MATLAB/Abaqus peak differences are 1.87–4.24%. Table 5 and Figure 3 report median costs and observed ranges from 90 runs; complete phase, memory and residual records accompany the code. All saved MATLAB numerical arrays and Abaqus response CSVs are identical across repeats within each setting.

A separate small/coarse CPU1 VTune software-sampling run reproduces the same mesh, boundary sets and 2,001 response rows exactly. It identifies 2.03 s of self-time estimates in the user-subroutine library and 3.81 s in four named assembly routines. These exclude callees and unresolved symbols; profiling effects and timer scopes prevent interpreting them as complete material/assembly wall times. Function and call-stack exports accompany the code; these estimates do not enter the performance table.

Benchmark 2: 3D Nooru-Mohamed Tests

The Nooru-Mohamed specimen is a square double edge notched concrete panel, $200 \times 200 \times 50$ mm. The illustrative mesh uses two 20 mm-deep, 5 mm-wide horizontal notches from opposite edges;

Table 5: Size/mesh computing study: median time in seconds from three sequential observations per setting. MATLAB covers the load loop; Abaqus covers analysis and output. The observed ranges are shown in Figure 3. GPU denotes the hybrid backend.

Size	Mesh	DOFs	MAT 1	MAT 8	MAT GPU	ABA 1	ABA 8
Small	coarse	3,772	31.3	33.8	38.5	158	234
Small	medium	14,686	131.2	144.0	146.1	832	403
Small	fine	25,534	232.3	251.8	255.0	1667	691
Large	coarse	14,686	127.6	140.0	147.0	819	398
Large	medium	56,954	598.9	639.6	626.4	4380	1586
Large	fine	101,088	1155.5	1206.0	1193.9	8548	2947

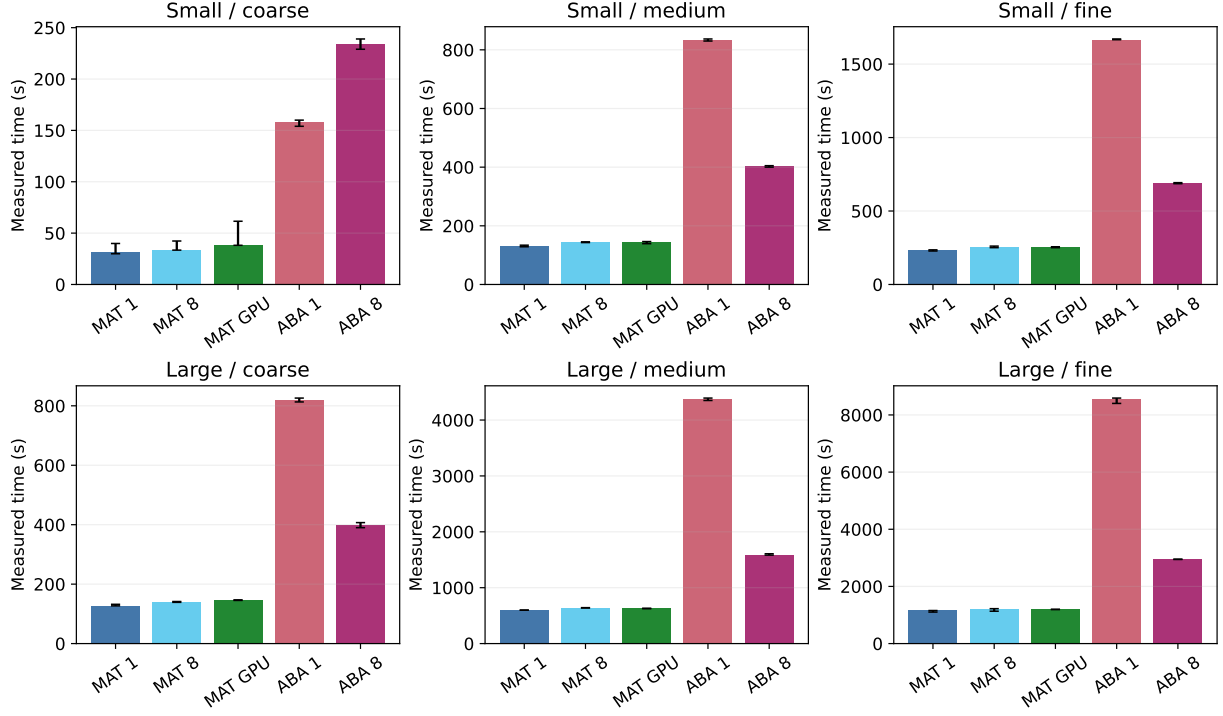


Figure 3: Thirty size/mesh/computing configurations with three observations each. Bars show medians; whiskers show the observed minimum and maximum. MATLAB bars cover the load loop; Abaqus bars cover analysis and output. CPU8 denotes a MATLAB library-thread limit or Abaqus SMP; GPU denotes element/damage work with CPU sparse factorization. These workstation observations establish no cross-program speed ranking or GPU speed advantage.

the published specimen has 25 mm-deep notches [8]. The central ligament between the notch tips experiences a mixed mode stress state. The illustrative proportional-displacement case prescribes a vertical (mode I) displacement on the top edge and a proportional horizontal (shear) displacement on the side edges, with a shear to tension ratio $\gamma = 0.6$ and the bottom edge held fixed, ramped together under displacement control over 900 increments to a top displacement of 0.5 mm. The material is concrete with $E = 29,000$ MPa, $\nu = 0.20$, $f_t = 3.0$ MPa, $G_F = 0.110$ N/mm, and $k = f_c/f_t = 10$ (so $f_c = 30$ MPa).

The proportional-loading damage panels illustrate the workflow. Its recorded shear and normal reaction curves are archived as diagnostics: 894 of 900 increments exceed the 10^{-6} equilibrium tolerance, so these histories do not support a quantitative mixed-mode validation. A strict rerun rejects unconverged trials and stops at target-bisection exhaustion; its failure record is preserved. The option name `4c` selects proportional displacement control, not the published sequential experimental loading. A separate pure-tension comparison uses specimen 47-05 with two glued platens (Figure 3.16, printed page 45 of [8]). Its load-displacement curve is manually digitized; estimated reading uncertainty is ± 0.00102 mm and ± 164 N. This quantitative case uses the published 25 mm-deep, 5 mm-wide notches, with 35,917, 94,525 and 144,791 TET4 elements (21,828, 54,540 and 84,774 DOFs). Global/notch mesh seeds are 12/4, 9/3 and 6/2 mm. Material parameters above are retained without fitting. Bottom vertical displacements are fixed, the top platen has a common unknown vertical displacement, and minimal lateral pins remove rigid modes. The mean displacement of four interpolated 65 mm vertical gauges at $x = 30/170$ mm on both faces controls the load history. An augmented Newton system solves this constraint with nodal equilibrium; a numerical constitutive tangent differentiates the same damage and projected-width update. The relative equilibrium tolerance is 10^{-6} ; unconverged trials are rejected and their target intervals bisected without committing damage. The initial 600 target intervals to 0.2 mm produce 608, 606 and 601 accepted increments after eight, six and one rejected/bisected trials. Coarse, medium and fine peaks are 16.64, 16.40 and 16.53 kN, versus the digitized 19.85 kN: errors are -16.18% , -17.39% and -16.76% . The peak spread is 1.46% of their mean. On 1,001 uniform points over 0–0.2 mm, experimental curve RMS errors are 9.27%, 9.25% and 9.09% of experimental peak; medium-fine RMS difference is 0.35% of fine peak. Both peak and late softening tail are underpredicted. All accepted residuals meet 10^{-6} ; the maximum is 9.84×10^{-7} and maximum gauge error is 1.24×10^{-15} mm. The small mesh spread does not remove the experimental discrepancy or establish general mesh independence. Idealized platens, constitutive assumptions, digitization and untested increment convergence limit interpretation.

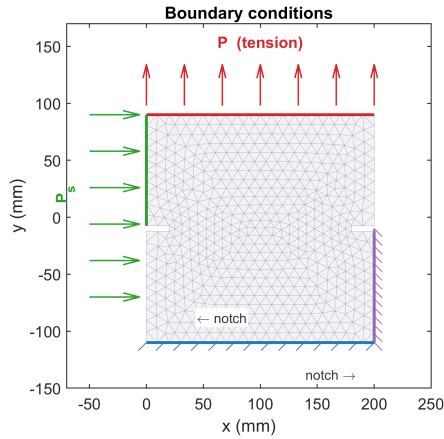
Local TET4 checks cover four edge lengths (0.5, 1, 2 and 4 mm) and eight tensile directions. All 32 cases pass equivalent-strain mapping, principal direction, projected width, finite-interval post-peak work and fixed-direction unloading checks. Compression mapping and eight rotating-direction unloading checks also pass. These verify the local implementation and irreversible history, independently of structural mesh convergence; their source-linked data accompany the code.

Benchmark 3: 3D Notched Beam Torsion

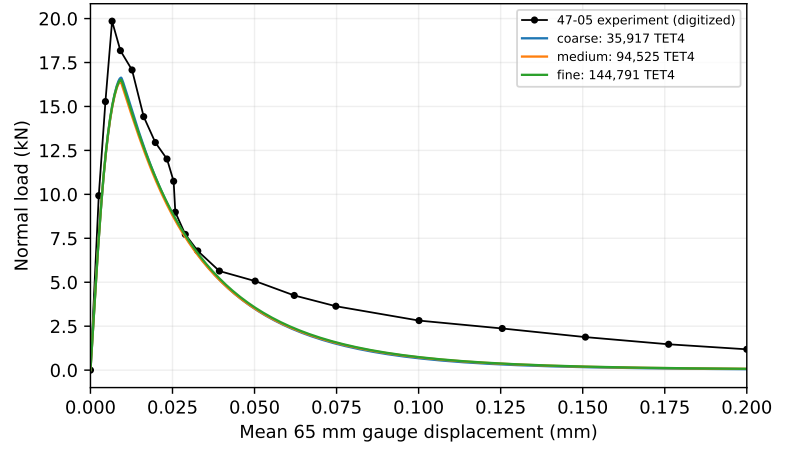
The torsion illustration uses a $400 \times 100 \times 100$ mm beam. The published Brokenshire test uses steel collars with three supported arms and one loaded arm [9, 10, 11]. Its measured response is applied load versus clip-gauge CMOD. The rotation-controlled model illustrated here uses different end constraints; a load-CMOD validation therefore requires a model of the collar and gauge configuration. The projected width follows the local principal direction, producing a nonplanar damage path whose mesh convergence is unverified.

In the model one end face is fully fixed while the opposite end face is given a prescribed rigid-body twist θ about the beam axis under rotation control, ramped to $\theta_{\text{end}} = 3.0 \times 10^{-3}$ rad over 140 increments; the reaction torque is recovered from the twisted end.

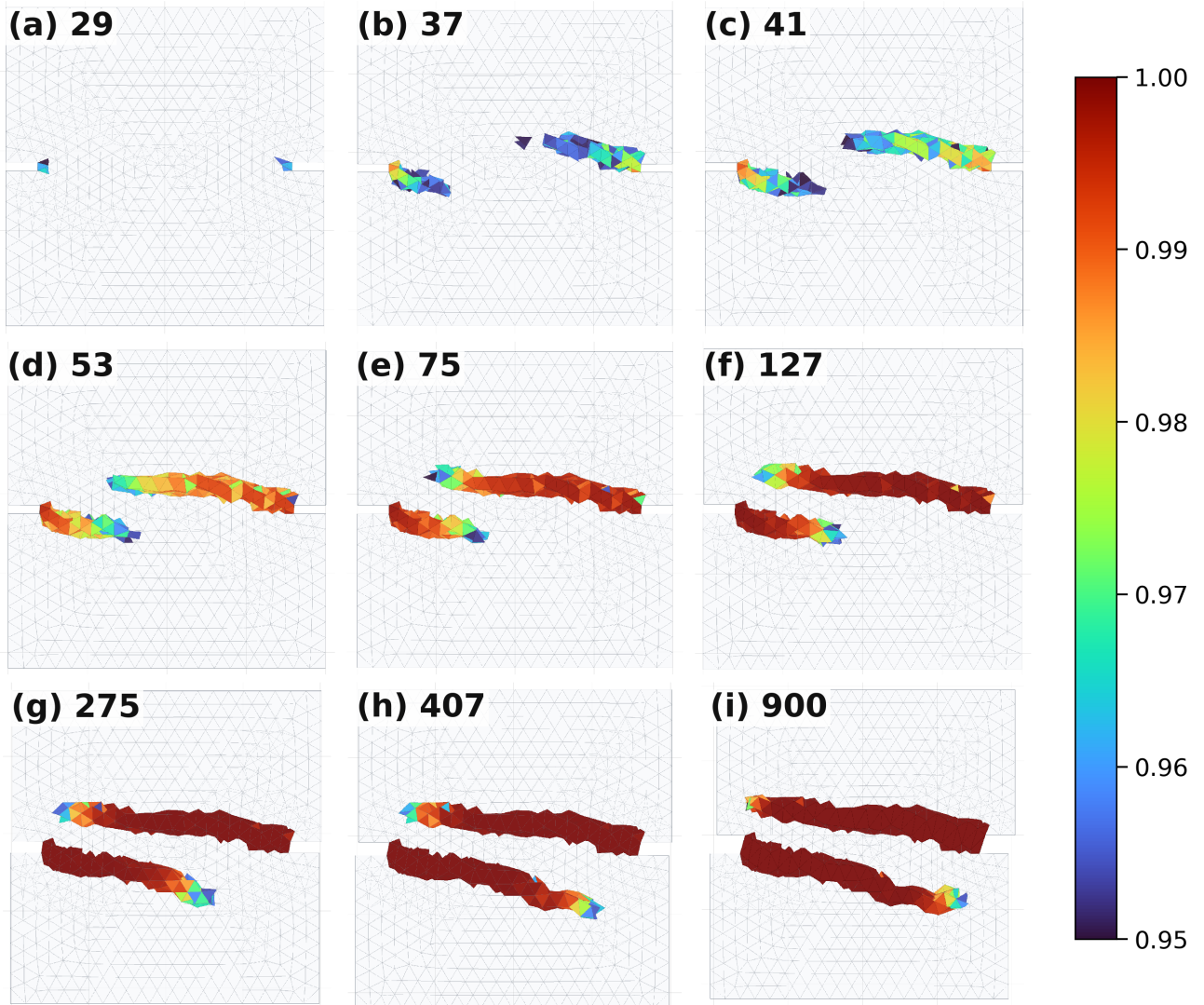
The material is plain concrete with $E = 35,000$ MPa, $\nu = 0.20$, $f_t = 3.0$ MPa, $G_F = 0.080$ N/mm, $\kappa_0 = f_t/E = 8.57 \times 10^{-5}$, and $k = f_c/f_t = 10$ (so $f_c = 30$ MPa), with the same exponential softening scaled element by element through the crack band length.



(a) 2D boundary conditions.

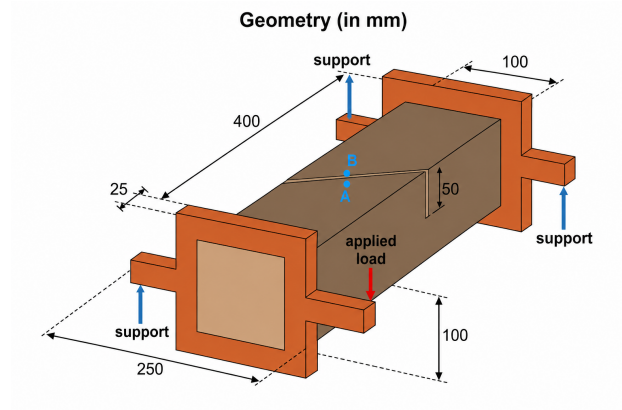


(b) Pure tension: published 25 mm notches.

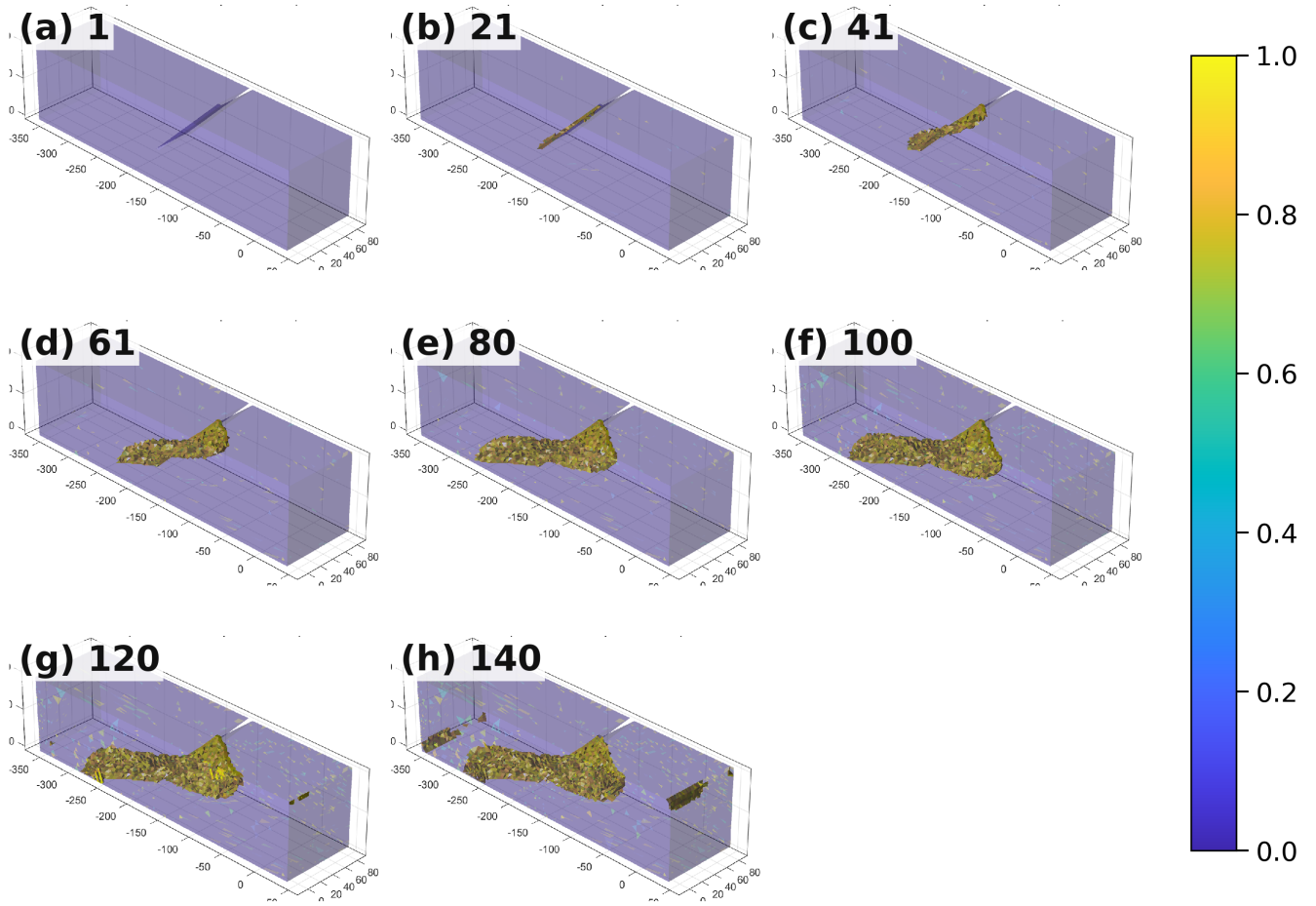


(c) Damage evolution from first localization at the notches to coalescence; panels (a)–(i), left to right and top to bottom, correspond to increments 29, 37, 41, 53, 75, 127, 275, 407, and 900.

Figure 4: Nooru-Mohamed cases: proportional-loading boundary conditions and damage panels; the load curves compare three separate pure-tension meshes with digitized specimen 47-05 data. Proportional-loading panels use 20 mm notches; the pure-tension curves use 25 mm notches. The controls differ. Mixed-mode damage panels are qualitative; their recorded reaction history contains equilibrium failures.



(a) Published collar loading arrangement; the illustrative model imposes an end-face twist.



(b) Torsion damage sequence.

Figure 5: Brokenshire torsion example. The 3D panels illustrate the workflow qualitatively; quantitative validation against a measured torque curve is outside this study.

The TET4 mesh carries a refined layer around the notch front. The damage update is under-relaxed with a factor of 0.35 to keep the staggered fixed point iteration stable through the steep post-peak branch, and each linear solve reuses a single AMD ordering with a Cholesky factorization.

This case is a qualitative demonstration of direction-dependent crack-band regularization under nonplanar, torsion-driven crack growth: the predicted damage band nucleates at the notch front and spirals toward the loaded corner (Figure 5), qualitatively similar to the published fracture description [9]. The available torsion history does not provide a valid experimental CMOD channel; it supports the qualitative damage illustration only.

Controlled mesh sensitivity and computational scaling

A controlled 2D mesh family contains 14,313, 25,042, and 56,216 CPS3 elements, corresponding to 14,686, 25,534, and 56,954 DOFs. The notched-beam geometry and material properties are unchanged. Additional partitions place the supports exactly at $x = 50$ and 300 mm and define a fixed 6.25 mm loading strip at the top centre; all nodes on that strip receive the same displacement. The 50 mm-wide central zone containing the notch is refined across the beam height. Its nominal seeds are 1.25, 0.9375, and 0.625 mm; the outer seeds are five times larger. This concentrates elements near the expected damage path while retaining a coarser outer region. In all three main Oliver runs, elements with $\omega \geq 0.95$ at peak load have centroids inside this refined region, at heights 20.2–37.6 mm above the beam bottom. This checks the chosen refinement location, without asserting convergence of the damage-band geometry. These boundary sets and each complete Abaqus connectivity are exported directly to MATLAB; mesh-file hashes are checked after Abaqus rebuilds. This family is distinct from the single-mesh increment study. Both programs use one computational thread/CPU.

The main MATLAB runs impose -0.1 mm in 2,000 increments ($|\Delta u| = 5 \times 10^{-5}$ mm). The regularized runs update the Oliver projected width; the control runs hold the softening calibration width at $h_{\text{ref}} = 1.25$ mm on every mesh. This control uses one fixed stress–strain law and disables element-size compensation. A second complete MATLAB family halves $|\Delta u|$ on all three meshes for both calibrations, using 4,000 increments. The Abaqus jobs use the same maximum displacement and increment cap, with automatic cutbacks available; their accepted counts are reported.

For each MATLAB increment, the undamaged energy density is $Y_e = \frac{1}{2} \boldsymbol{\varepsilon}_e^T \mathbf{D} \boldsymbol{\varepsilon}_e$. Accumulated damage dissipation is evaluated by the positive end-of-step quadrature

$$\mathcal{D}_N = \sum_{n=1}^N \sum_e Y_e^n (\omega_e^n - \omega_e^{n-1}) V_e. \quad (4)$$

External work uses trapezoidal integration of reaction versus prescribed displacement; stored energy is $\sum_e (1 - \omega_e) Y_e V_e$. The discrepancy between external work and stored energy plus $\mathcal{D}_N$ is retained as a numerical diagnostic. Energy comparisons use a common CMOD of 0.10 mm and represent partial structural dissipation, not a measurement of complete fracture energy.

Figure 6 compares the two softening calibrations. Coarse, medium and fine load–CMOD curves are overlaid separately for Oliver MATLAB, fixed-law MATLAB and regularized Abaqus. The three MATLAB/Abaqus pairs are also plotted in the code archive; their full responses reveal peak and post-peak differences that timings cannot assess. Across the three main runs, peak-load spread, defined as $(\max - \min)/\text{mean}$, is 6.19% with Oliver regularization and 6.78% with the fixed law. The corresponding dissipation spreads are 6.86% and 18.21% (Table 6). The comparison tests element-size compensation for this mesh family; remaining sensitivity does not support complete mesh independence.

At 4,000 increments on every mesh, peak-load spreads are 5.98% (Oliver) and 8.05% (fixed law); dissipation spreads are 6.44% and 20.76%, respectively. The reduction in dissipation sensitivity therefore persists at the smaller increment. Halving the displacement increment gives coarse fixed (−0.96% peak, −1.27% dissipation), coarse oliver (−0.89% peak, −0.99% dissipation), medium fixed (−1.85% peak, −2.55% dissipation), medium oliver (−1.37% peak, −1.53% dissipation), fine fixed (−2.20% peak, −3.74% dissipation), fine oliver (−1.31% peak, −1.42% dissipation). These finite-increment

Table 6: Controlled mesh study, 2,000 MATLAB increments. Partial damage dissipation $\mathcal{D}$ is evaluated at CMOD 0.10 mm. Subscripts O, F, and A denote Oliver MATLAB, fixed-law MATLAB, and regularized Abaqus runs.

Elements	DOFs	P_O (kN)	P_F (kN)	P_A (kN)	$\mathcal{D}_O$ (N mm)	$\mathcal{D}_F$ (N mm)
14,313	14,686	4.194	4.025	4.117	140.08	132.13
25,042	25,534	4.205	3.808	4.087	140.93	117.25
56,216	56,954	4.459	3.763	4.339	149.94	110.30

checks quantify sensitivity; they do not establish asymptotic convergence. At CMOD 0.10 mm, the absolute energy-balance discrepancy is 0.42–1.00% of external work in the main runs and 0.21–0.48% in the refined runs. The fine Oliver peak post-damage relative residual decreases from 0.0052 to 0.0029. The residual and energy discrepancy remain explicit limits of the sequential update.

MATLAB peak loads differ from the corresponding Abaqus peaks by +1.87%, +2.88%, +2.78%, respectively, relative to Abaqus. The completed Abaqus jobs report coarse: 2 cutbacks and 5,152 decompositions, medium: 0 cutbacks and 6,153 decompositions, fine: 0 cutbacks and 7,502 decompositions. Alternate force-tolerance acceptances are 4, 3, 5, respectively. The rounded analysis CPU totals from the final `.dat` summaries are 531 s, 1100 s, 2550 s, respectively. CPU and elapsed timers have different scopes; these totals do not isolate material or assembly work. Their wall time outside the sparse-solver timer is 794.2 s, 1585.6 s, 3979.7 s, respectively. It includes unseparated element/UMAT, assembly, convergence, output, and overhead; no material-time attribution is inferred. The fine-mesh response overlay is a code-to-code check with different equilibrium algorithms. Machine-readable histories, exact meshes, input decks, diagnostics, and analysis scripts accompany the study.

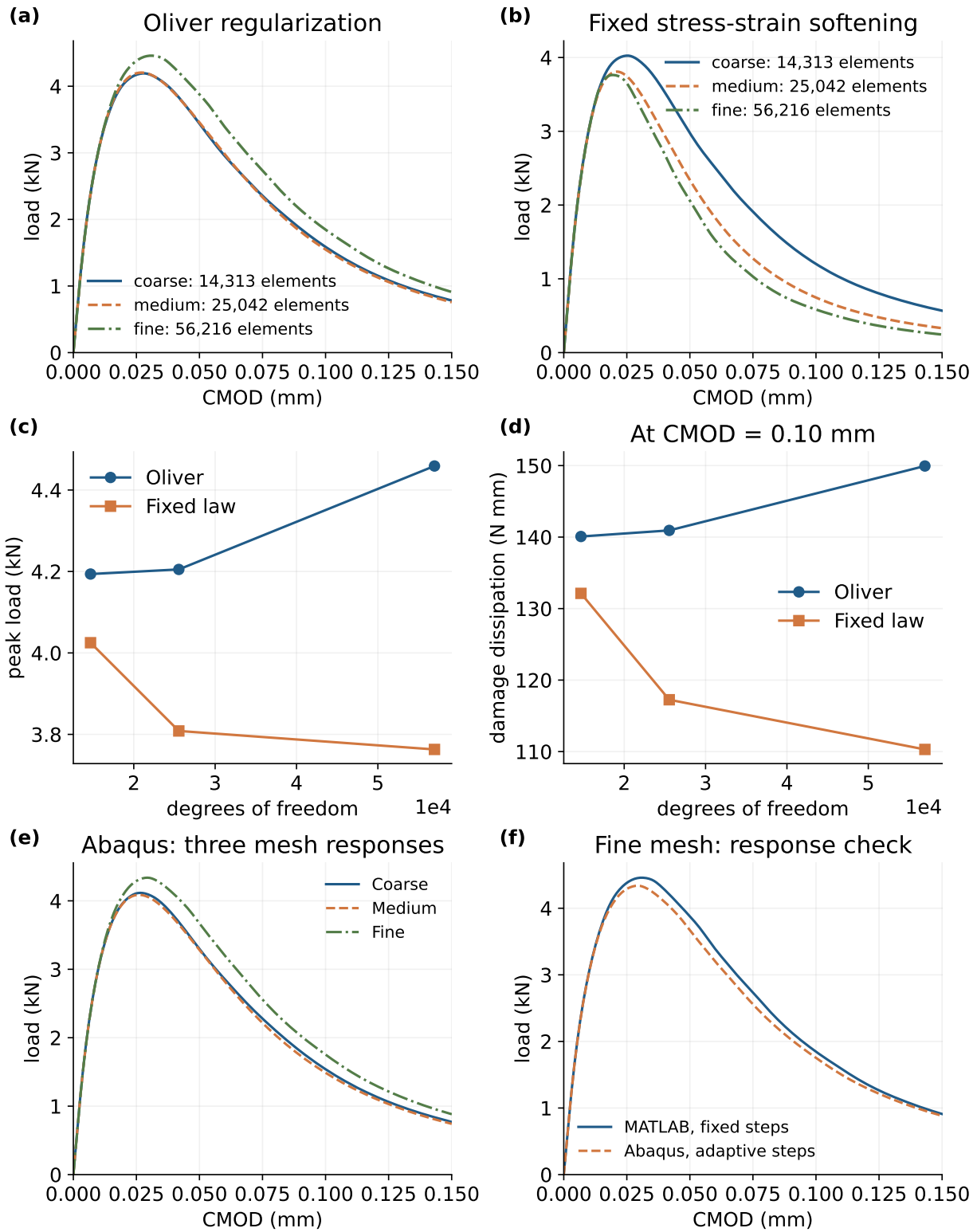


Figure 6: Controlled 2D study. (a,b) Oliver and fixed-law responses; (c,d) peak load and partial damage dissipation versus DOFs; (e) regularized Abaqus responses on the three meshes; (f) fine-mesh MATLAB/Abaqus responses. Main MATLAB curves use 2,000 increments to -0.1 mm; Abaqus uses the same maximum increment with adaptive cutbacks. Equilibrium algorithms differ. Timing comparisons appear separately in Fig. 3.

Impact

The main value of **FRACMATH** is that users can trace a complete damage calculation through mesh import, element matrices, projected bandwidth, constitutive history, sparse solve, and plotted response. Material-point tests verify strain mapping and softening calibration; the recorded timing scopes distinguish assembly from sparse factorization. The 2D and 3D examples provide starting cases for method development and classroom use. The supplied scripts permit the figures to be regenerated from the associated data. Users can investigate how mesh and increment size affect damage dissipation, and whether moving element work to a GPU helps when sparse factorization remains on the CPU. The fixed-law control, residual histories and measured component costs make these questions testable. The beginner entry script and array guide support classroom inspection. Independent adoption, download statistics and commercial use are not established by this study.

Limitations

The model is local, isotropic, quasistatic, and intended for monotonic loading. A single damage scalar does not represent crack closure or anisotropic degradation. The material-point tests verify softening calibration. The controlled 2D family tests mesh sensitivity for one geometry and mesh construction; it does not establish general mesh or orientation independence. The three-mesh 3D pure-tension comparison uses published 25 mm notches and idealized platens; peaks remain 16.18–17.39% below the digitized experiment. General 3D mesh independence is not established. Mixed-mode and torsion images remain qualitative. The 2D sequential update leaves a nonzero post-damage equilibrium residual; the increment and refinement studies quantify its effect. A fully coupled outer equilibrium iteration is future work. The hybrid GPU backend applies to 2D T3 elements and retains CPU sparse factorization. Three observations per configuration on one workstation describe observed timing variation, without establishing population confidence intervals. Different equilibrium algorithms and timing scopes prevent an accuracy-matched cross-program speed ranking.

Conclusions

FRACMATH is an inspectable MATLAB example of local damage with an Oliver projected crack-band width. The study presents constitutive checks, controlled 2D mesh and increment studies, structural energy diagnostics, and timing scopes that separate sparse factorization from assembly in MATLAB. A geometry-matched 3D pure-tension comparison quantifies peak and curve errors without fitting material parameters; local TET4 checks verify projected width and irreversible history. Ninety observations in the hybrid GPU and SMP study measure hardware-dependent costs on matched meshes. The mixed-mode and torsion cases remain qualitative, and complete Abaqus material/assembly wall costs remain unmeasured; separate sampling identifies user-library and assembly self work.

Data availability

The MATLAB and UMAT sources, 2D/3D meshes, material-point tests, machine-readable response histories, timing observations and figure scripts are available in the [FRACMATH GitHub repository](#) [12]. The code and study data are identified by the immutable Git commit listed in Table 1.

CRediT authorship contribution statement

Jaykumar Mavani: Methodology, Software, Validation, Visualization, Writing – original draft.
Madura Pathirage: Conceptualization, Supervision, Funding acquisition.

Acknowledgements

The authors acknowledge support from the National Aeronautics and Space Administration (NASA), and the New Mexico Space Grant Consortium under grant 80NSSC22M0044.

Declaration of competing interest

The authors declare no conflict of interest.

Declaration of AI-assisted technologies

OpenAI Codex was used to assist with manuscript wording, formatting, and implementation support. The scientific claims, equations, code, numerical results, figures, and references are the responsibility of the authors.

References

- [1] J. H. P. de Vree, W. A. M. Brekelmans, M. A. J. van Gils, Comparison of nonlocal approaches in continuum damage mechanics, *Computers & Structures* 55 (4) (1995) 581–588. doi:10.1016/0045-7949(94)00501-S.
- [2] J. Oliver, A consistent characteristic length for smeared cracking models, *International Journal for Numerical Methods in Engineering* 28 (2) (1989) 461–474. doi:10.1002/nme.1620280214.
- [3] Z. P. Bažant, B. H. Oh, Crack band theory for fracture of concrete, *Matériaux et Construction* 16 (3) (1983) 155–177. doi:10.1007/BF02486267.
- [4] Dassault Systèmes Simulia Corp., *Abaqus/Standard User’s Manual*, Version 2024, Providence, RI, USA, 2024.
- [5] B. Patzák, Z. Bittnar, Design of object-oriented finite element code, *Advances in Engineering Software* 32 (10–11) (2001) 759–767. doi:10.1016/S0965-9978(01)00027-8.
- [6] P.-E. Austrell, O. Dahlblom, J. Lindemann, A. Olsson, K.-G. Olsson, K. Persson, H. Petersson, M. Ristinmaa, G. Sandberg, P.-A. Wernberg, *CALFEM—A Finite Element Toolbox*, Version 3.4, Lund University, Lund, Sweden, 2004.
- [7] D. Grégoire, L. B. Rojas-Solano, G. Pijaudier-Cabot, Failure and size effect for notched and unnotched concrete beams, *International Journal for Numerical and Analytical Methods in Geomechanics* 37 (10) (2013) 1434–1452. doi:10.1002/nag.2180.
- [8] M. B. Nooru-Mohamed, Mixed-mode fracture of concrete: an experimental approach, Ph.D. thesis, Delft University of Technology, 1992. URL: <https://resolver.tudelft.nl/uuid:a6a773f1-dacd-4598-aa6a-960dddf71117>.
- [9] D. R. Brokenshire, A study of torsion fracture tests, Ph.D. thesis, Cardiff University, Cardiff, United Kingdom, 1996.
- [10] A. D. Jefferson, B. I. G. Barr, T. Bennett, S. C. Hee, Three dimensional finite element simulations of fracture tests using the Craft concrete model, *Computers and Concrete* 1 (3) (2004) 261–284. doi:10.12989/cac.2004.1.3.261.
- [11] A. D. Jefferson, T. Bennett, Benchmark problems for assessing a combined micro-mechanical continuum based model for concrete, COMPLAS VIII, Barcelona, 2005. URL: <https://congress.cimne.com/complas05/admin/Files/FilePaper/p334.pdf>.
- [12] J. Mavani, M. Pathirage, FRACMATH, Version 1.1.1, Zenodo, 2026. doi:10.5281/zenodo.23138595. Git commit f4d208e. URL: <https://github.com/Jaykumar9033/FRACMATH/tree/f4d208ecf45b6a4e4d4530e5f71fbf8bcb9f0fe0>.